

Issue # 7: Numerator Data - Trends and Event Studies 2021-2022 (weighted)

Ron Huang

October 5, 2025

This Write-up contains tables and plots for trend and event study analyses of the Numerator data. The original user-trip-item level data is transformed to user- month level data, with variables summarized at the monthly level for each user. SNAP and Non-SNAP treatment is determined by SNAP participation rate of each user over the periods of the data (>0.5 on SNAP; ≤ 0.5 Non-SNAP). We use the demographic weight in the data for analysis. The weight is relative to static panel cohort (each cohort constituted of households with the same 12-month static window). The sum of weights in each cohort equals to the count of total household. There are two cohorts for these analyses, one stars at Jan 2021 and one at Jan 2022. Not adjusted for inflation.

1. Figures: Monthly average of ...

- level/log total spending winsorized
- level/log Food spending winsorized
- level/log Food on EBT winsorized
- level/log Soda spending winsorized
- level/log Soda on EBT winsorized
- prop. food spending
- prop. food on EBT
- prop. soda
- prop. soda on EBT
- level/log soda spending by SNAP/Non-SNAP
- level/log soda spending by state $\times$ SNAP

2. Figures: Event Studies

- level/log soda spending winsorized
- level/log soda spending by SNAP $\times$ post (conditional on treated states)
- level/log soda spending DDD

3. Tables

- TWFE/DDD Model Estimates

Table 1: User-Month Panel - Demographics

	Treated States	Mean (SD) Control States	Total	Min	Max
Income	83511.90 (60560.08)	89083.98 (64848.46)	88182.36 (64206.78)	10000.00	275000.00
Household Size	2.78 (1.43)	2.84 (1.45)	2.83 (1.45)	1.00	7.00
Age	51.53 (12.51)	49.83 (12.41)	50.11 (12.44)	22.50	67.50
Married	0.59 (0.49)	0.57 (0.50)	0.57 (0.49)	0.00	1.00
Education (years)	13.86 (1.33)	13.91 (1.36)	13.90 (1.36)	10.00	16.00
White	0.76 (0.43)	0.69 (0.46)	0.70 (0.46)	0.00	1.00
Black	0.08 (0.28)	0.11 (0.31)	0.10 (0.30)	0.00	1.00
Asian	0.03 (0.16)	0.08 (0.27)	0.07 (0.25)	0.00	1.00
Hispanic	0.12 (0.32)	0.11 (0.31)	0.11 (0.31)	0.00	1.00
Ever Child	0.26 (0.44)	0.28 (0.45)	0.27 (0.45)	0.00	1.00
Number of States	9	48	57		
Number of Users	32,037	164,389	196,426		
Average Number of Months per User	18.15	18.32	18.29		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 2: User-Month Panel - Spending

	Treated States	Mean (SD) Control States	Total	Min	Max
Food Spending	485.37 (656.40)	504.22 (724.16)	501.17 (713.67)	0.00	87740.50
Total Spending	1266.58 (1945.46)	1296.22 (1883.64)	1291.43 (1893.81)	0.00	301562.55
Prop. of Total Spending on Food	0.42 (0.19)	0.42 (0.19)	0.42 (0.19)	0.00	1.00
Prop. of Food Spending on FAH	0.91 (0.12)	0.91 (0.13)	0.91 (0.13)	0.00	1.00
Prop. of Food Spending on FAFH	0.09 (0.12)	0.09 (0.13)	0.09 (0.13)	0.00	1.00
Prop of Food on EBT	0.38 (0.28)	0.38 (0.28)	0.38 (0.28)	0.00	1.00
Prop. of EBT Spending on FAH	1.00 (0.02)	1.00 (0.03)	1.00 (0.03)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.02)	0.00 (0.03)	0.00 (0.03)	0.00	1.00
Ever SNAP	0.18 (0.38)	0.19 (0.39)	0.19 (0.39)	0.00	1.00
Number of States	9	48	57		
Number of Users	32,037	164,389	196,426		

Notes: Prop. of food spending is calculated out of total spending. Prop. of FAH/FAFH spending and food on SNAP is calculated out of total food spending. Prop. of FAH/FAFH on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 3: User-Month Panel - Soda Spending

	Treated States	Mean (SD) Control States	Total	Min	Max
Soda Spending	17.31 (38.13)	16.48 (43.86)	16.62 (42.98)	0.00	10453.12
Prop. of Food Spending on Soda	0.04 (0.06)	0.04 (0.05)	0.04 (0.05)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.89 (0.24)	0.89 (0.25)	0.89 (0.25)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.11 (0.24)	0.11 (0.25)	0.11 (0.25)	0.00	1.00
Prop of EBT Spending on Soda	0.05 (0.11)	0.05 (0.10)	0.05 (0.10)	0.00	1.00
Prop of Soda Spending on EBT	0.35 (0.37)	0.36 (0.38)	0.36 (0.38)	0.00	1.00
Number of Users	32,037	164,389	196,426		

Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 4: User-Month Panel - Demographics (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Income	53435.48 (51293.14)	95144.96 (64272.68)	88182.36 (64206.78)	10000.00	275000.00
Household Size	3.04 (1.64)	2.79 (1.40)	2.83 (1.45)	1.00	7.00
Age	49.60 (12.27)	50.21 (12.47)	50.11 (12.44)	22.50	67.50
Married	0.34 (0.47)	0.62 (0.49)	0.57 (0.49)	0.00	1.00
Education (years)	13.44 (1.41)	14.00 (1.32)	13.90 (1.36)	10.00	16.00
White	0.52 (0.50)	0.73 (0.44)	0.70 (0.46)	0.00	1.00
Black	0.21 (0.40)	0.08 (0.27)	0.10 (0.30)	0.00	1.00
Asian	0.05 (0.23)	0.07 (0.26)	0.07 (0.25)	0.00	1.00
Hispanic	0.19 (0.39)	0.10 (0.29)	0.11 (0.31)	0.00	1.00
Ever Child	0.35 (0.48)	0.26 (0.44)	0.27 (0.45)	0.00	1.00
Number of States	54	57	57		
Number of Users	33,982	162,444	196,426		
Average Number of Months per User	17.11	18.54	18.29		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 5: User-Month Panel - Spending (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Food Spending	577.10 (872.88)	485.96 (676.25)	501.17 (713.67)	0.00	87740.50
Total Spending	1348.20 (2005.08)	1280.05 (1870.51)	1291.43 (1893.81)	0.00	301562.55
Prop. of Total Spending on Food	0.46 (0.18)	0.42 (0.19)	0.42 (0.19)	0.00	1.00
Prop. of Food Spending on FAH	0.91 (0.12)	0.91 (0.13)	0.91 (0.13)	0.00	1.00
Prop. of Food Spending on FAFH	0.09 (0.12)	0.09 (0.13)	0.09 (0.13)	0.00	1.00
Prop of Food on EBT	0.44 (0.28)	0.24 (0.24)	0.38 (0.28)	0.00	1.00
Prop. of EBT Spending on FAH	1.00 (0.02)	1.00 (0.03)	1.00 (0.03)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.02)	0.00 (0.03)	0.00 (0.03)	0.00	1.00
Ever SNAP	0.81 (0.39)	0.06 (0.24)	0.19 (0.39)	0.00	1.00
Number of States	54	57	57		
Number of Users	33,982	162,444	196,426		

Notes: Prop. of food spending is calculated out of total spending. Prop. of FAH/FAFH spending and food on SNAP is calculated out of total food spending. Prop. of FAH/FAFH on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization. Food spending and total spending are higher for SNAP than Non-SNAP household, potentially reflecting the consequence of Numerator's monetary incentives. Household on SNAP, who tend to earn less income, tend to upload more receipt with larger values (incentives vary based on receipt value).

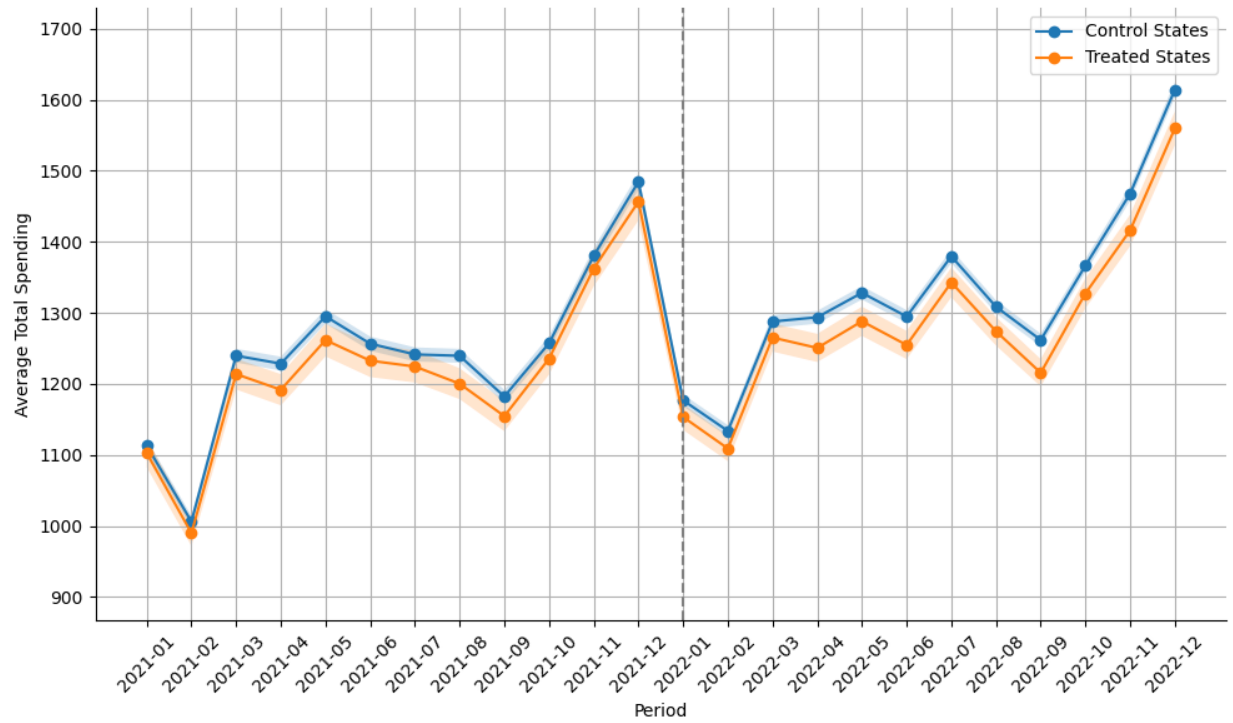
Table 6: User-Month Panel - Soda Spending (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Soda Spending	22.59 (48.62)	15.42 (41.66)	16.62 (42.98)	0.00	10453.12
Prop. of Food Spending on Soda	0.04 (0.05)	0.03 (0.05)	0.04 (0.05)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.91 (0.21)	0.89 (0.26)	0.89 (0.25)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.09 (0.21)	0.11 (0.26)	0.11 (0.25)	0.00	1.00
Prop of EBT Spending on Soda	0.05 (0.09)	0.05 (0.13)	0.05 (0.10)	0.00	1.00
Prop of Soda Spending on EBT	0.41 (0.38)	0.22 (0.32)	0.36 (0.38)	0.00	1.00
Number of Users	33,982	162,444	196,426		

Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Figure 1: Monthly Average of Total Spending

(a) level



(b) log

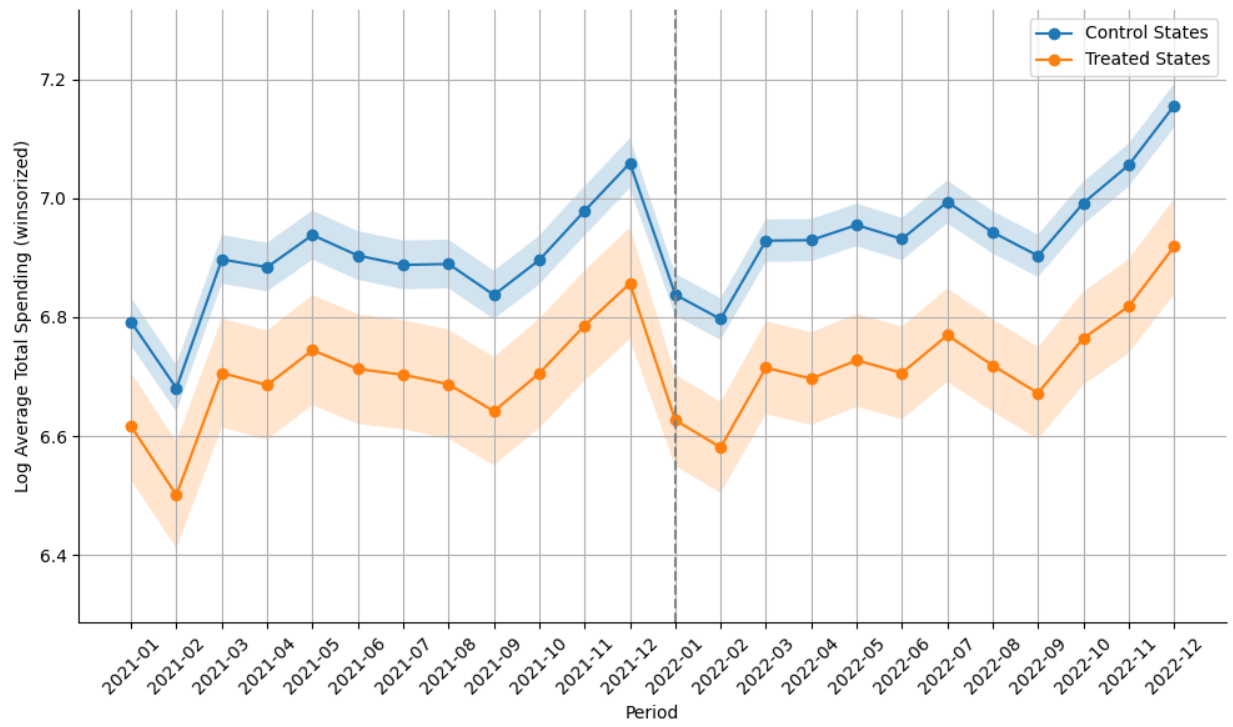
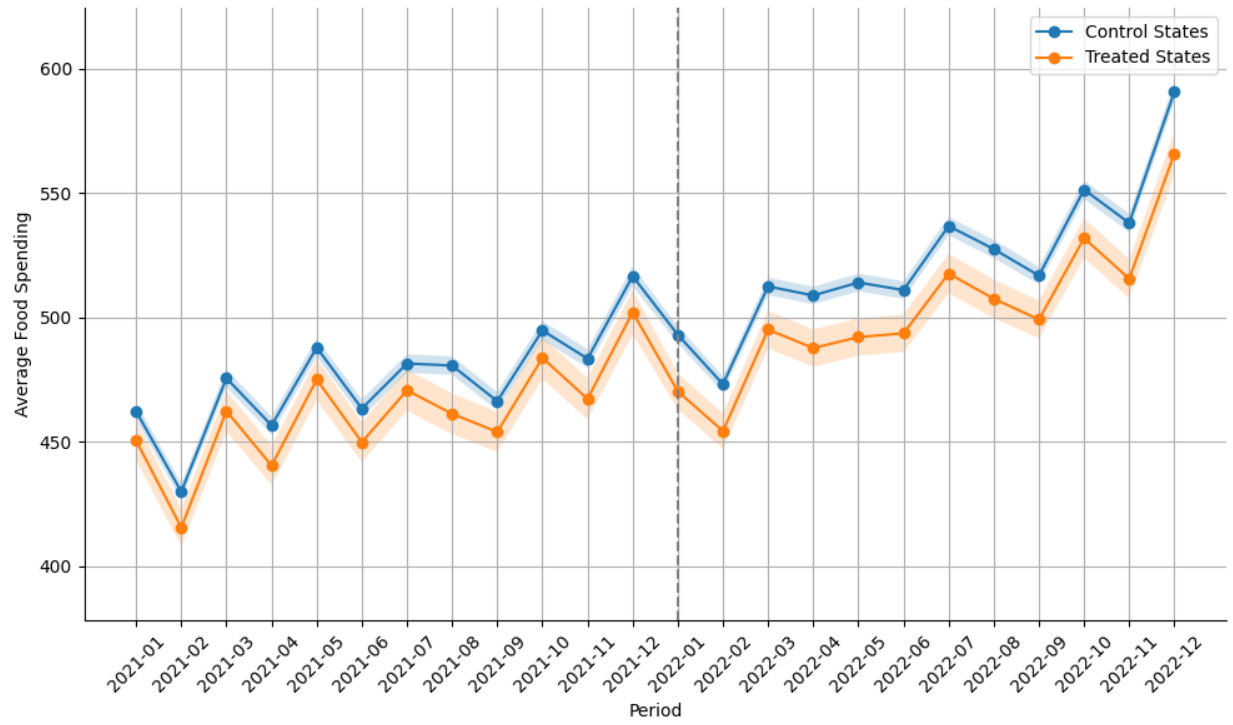


Figure 2: Monthly Average of Food Spending

(a) level



(b) log

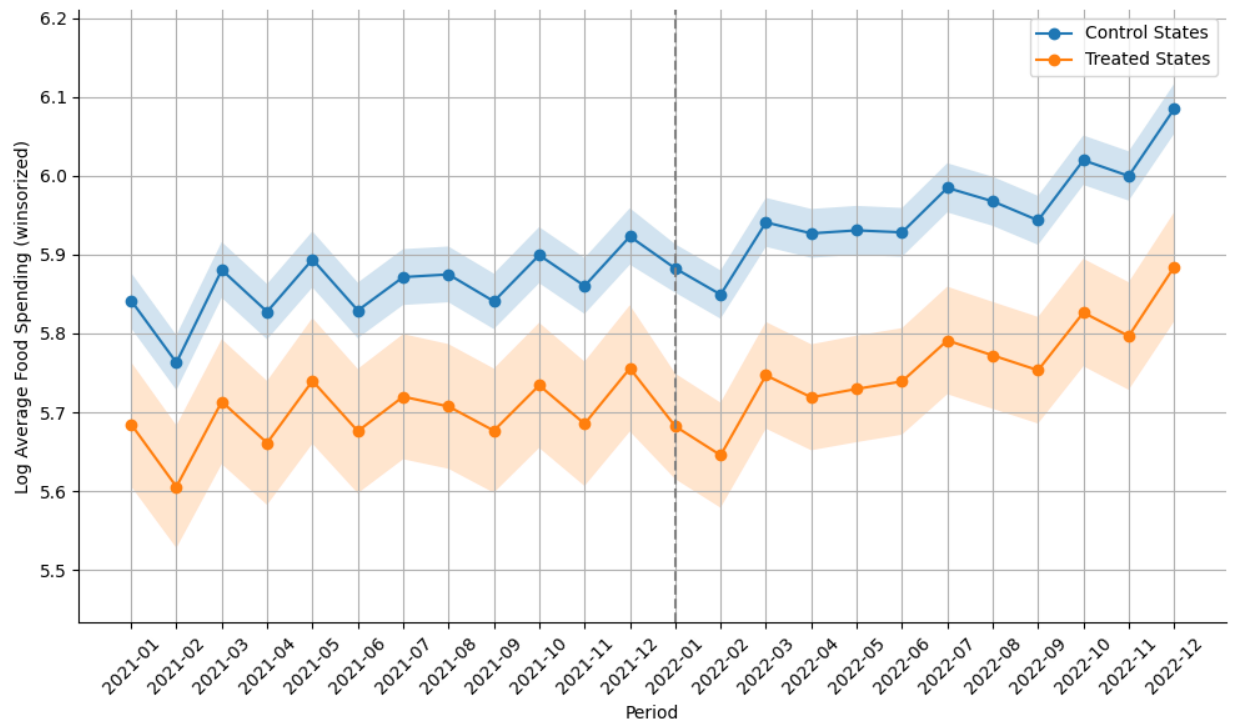
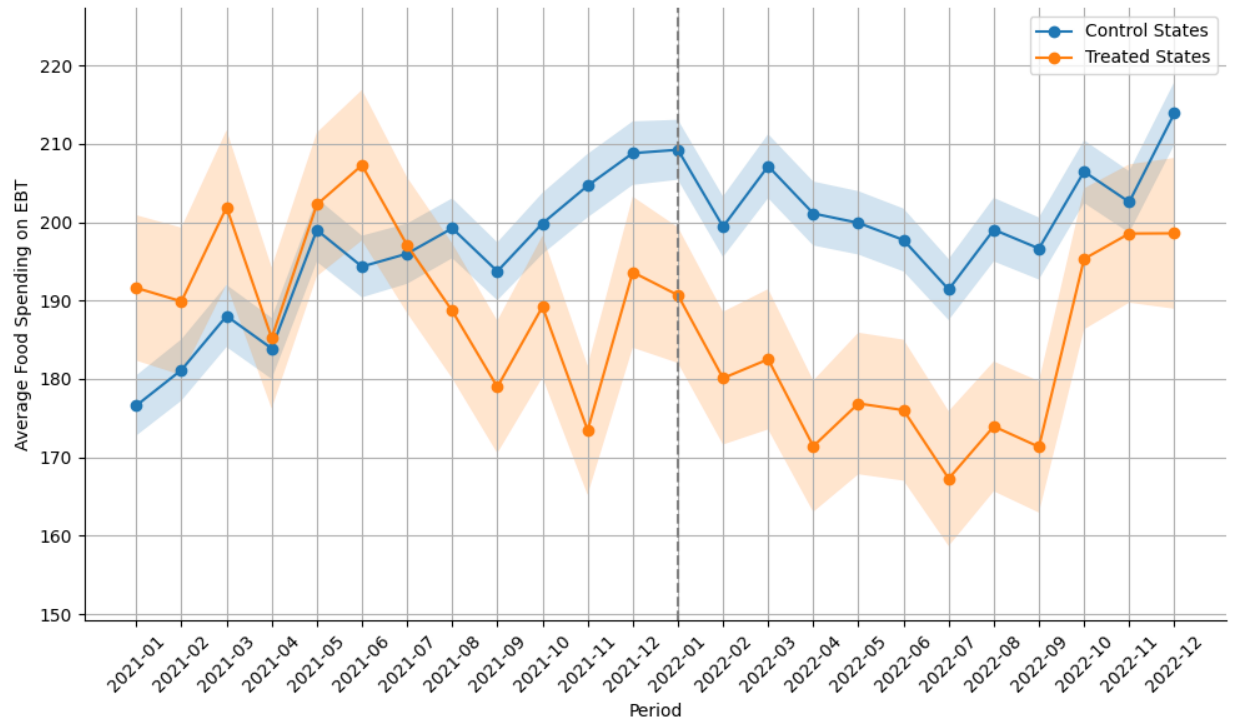


Figure 3: Monthly Average of Food Spending on EBT

(a) level



(b) log

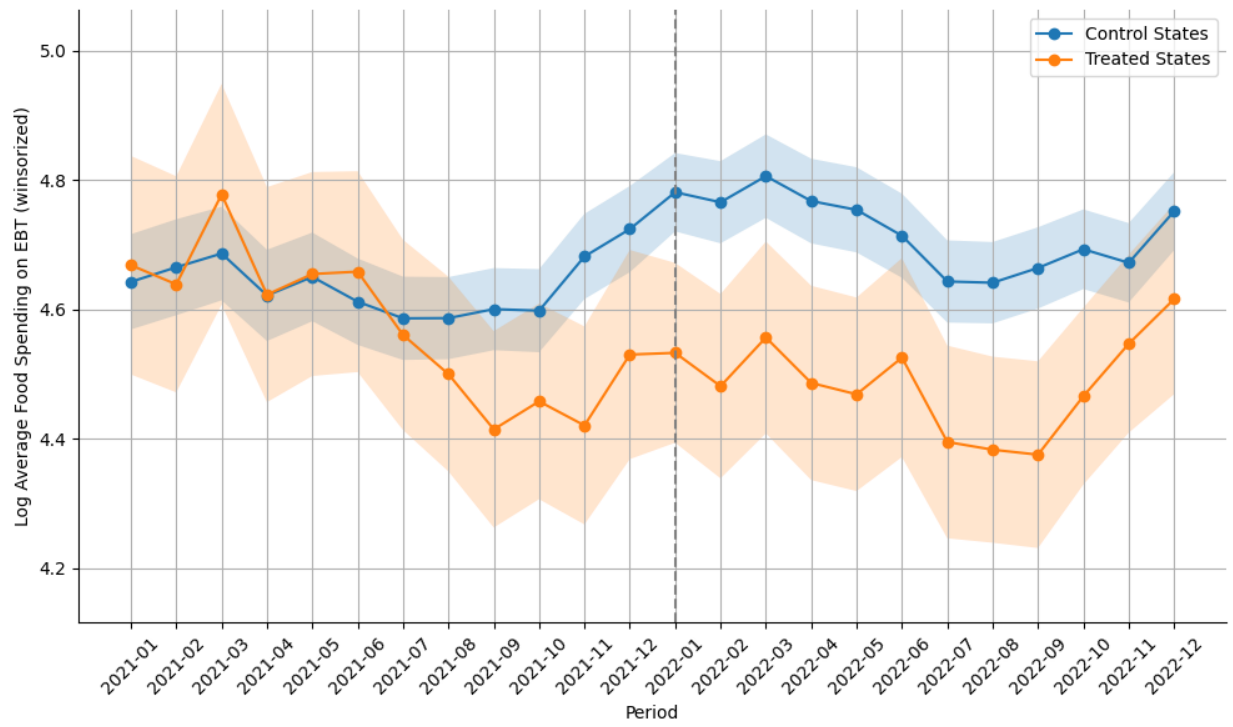
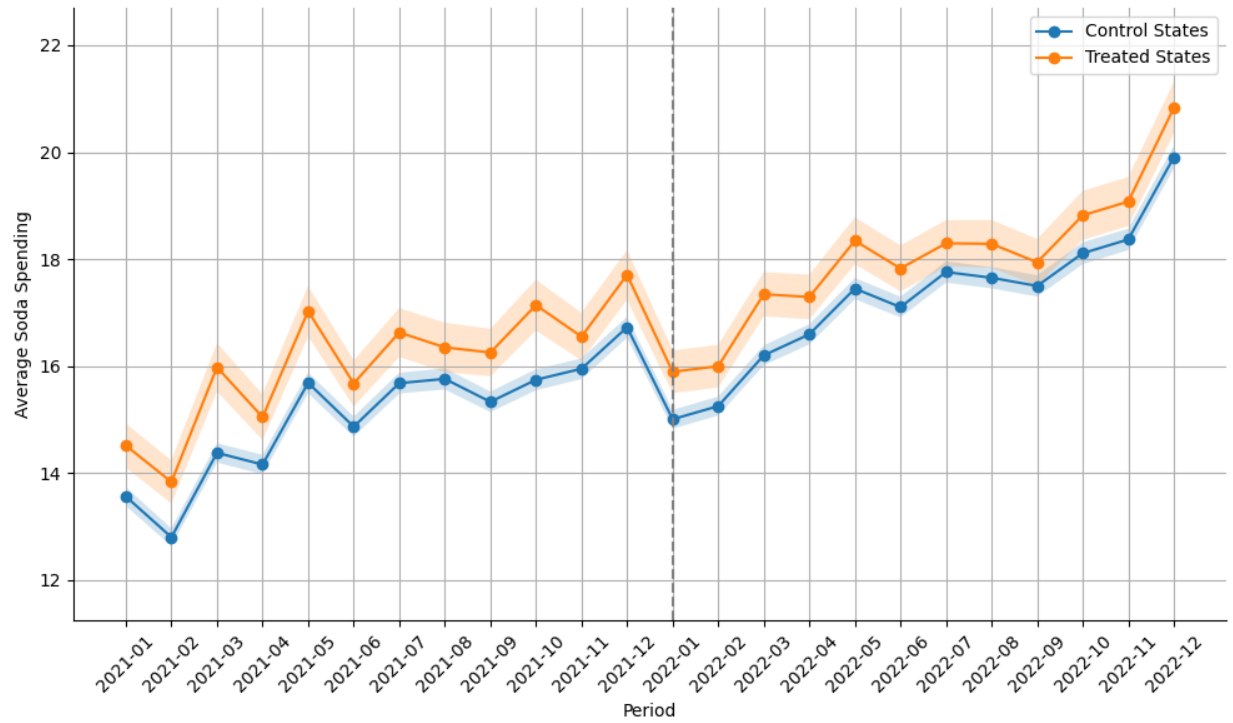


Figure 4: Monthly Average of Soda Spending

(a) level



(b) log

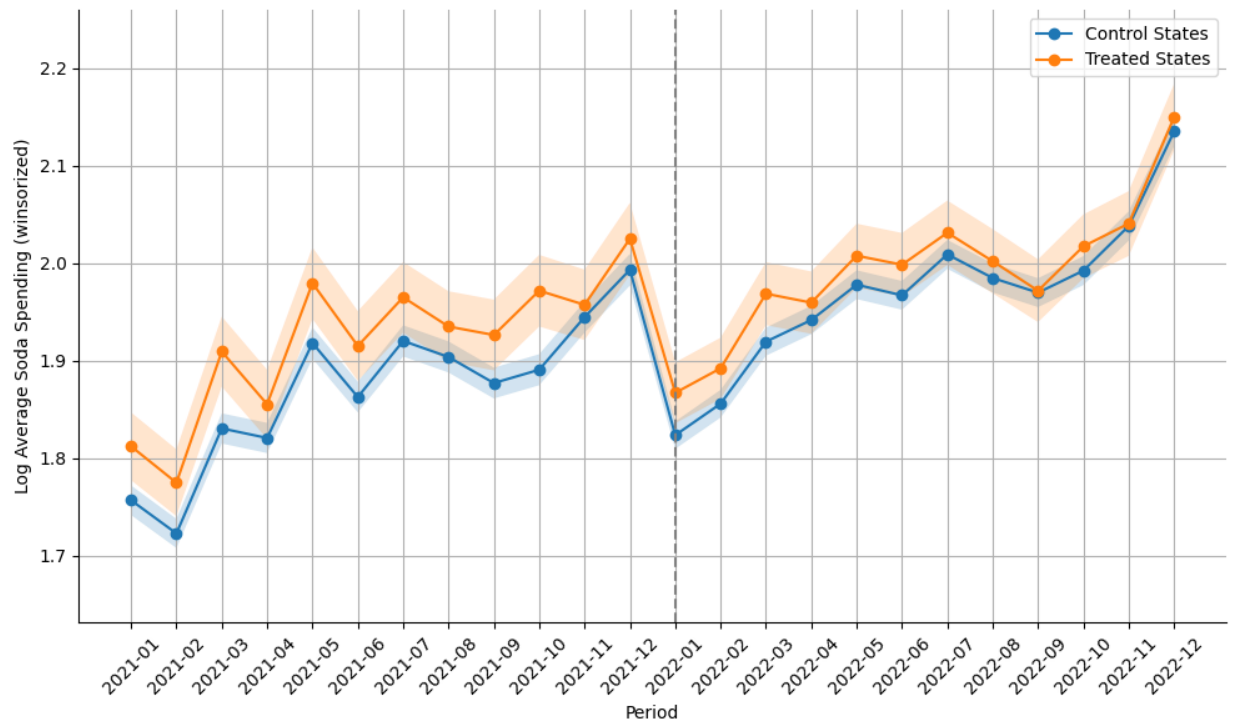
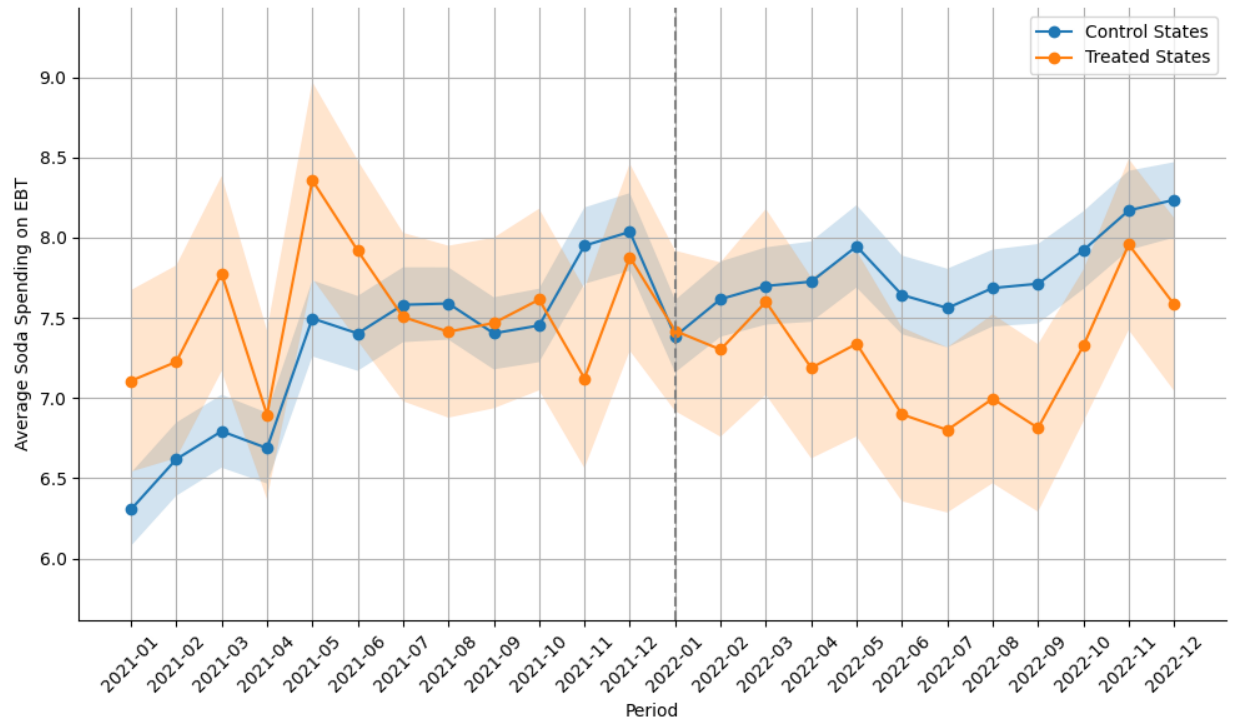


Figure 5: Monthly Average of Soda Spending on EBT

(a) level



(b) log

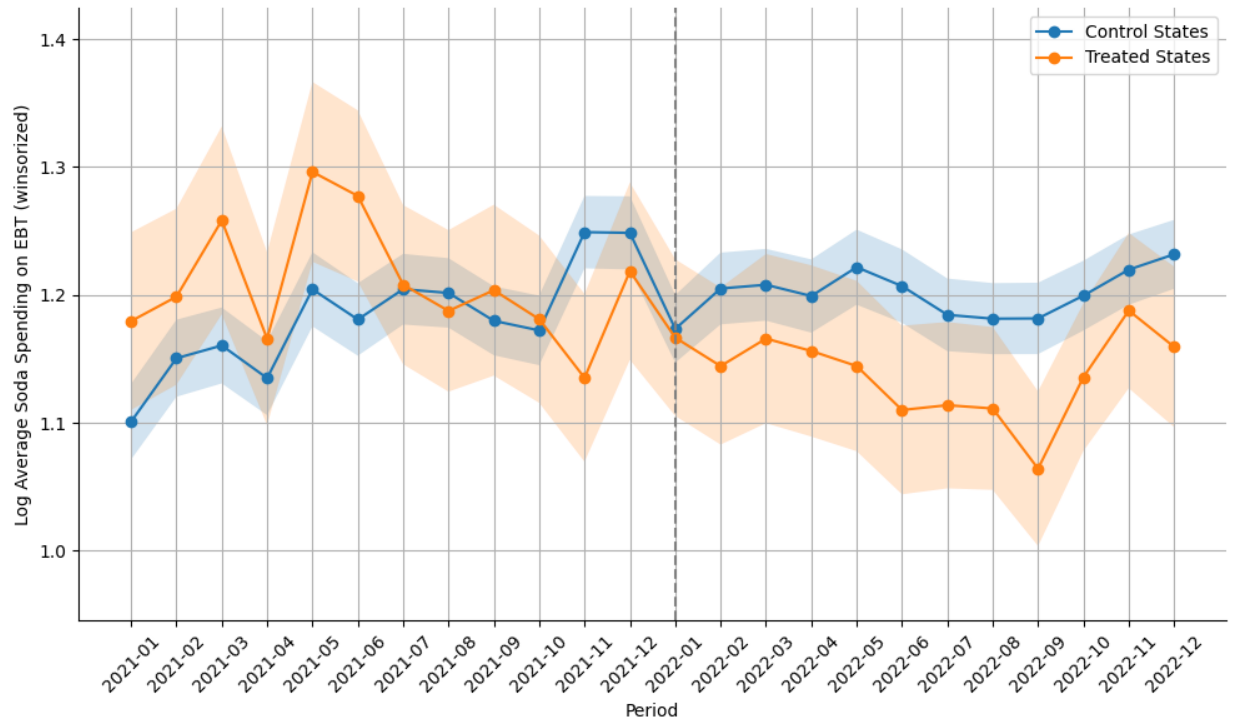
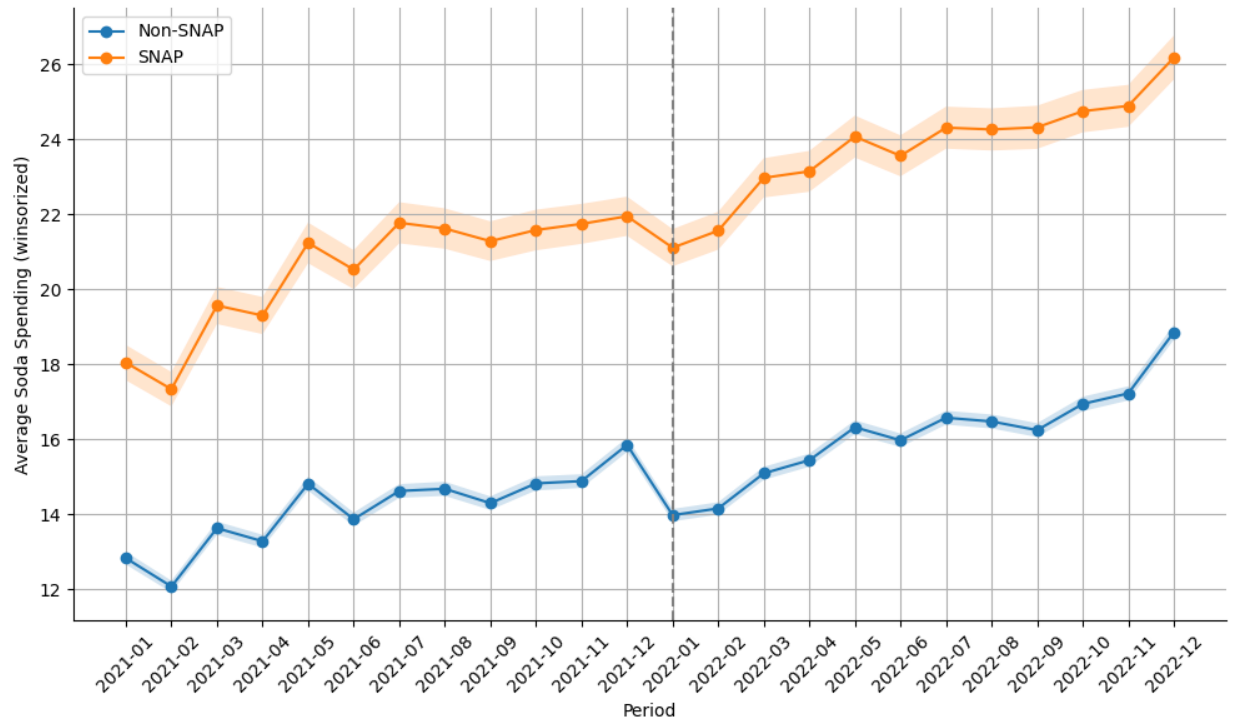


Figure 6: Monthly Average of Soda Spending by SNAP/Non-SNAP

(a) level



(b) log

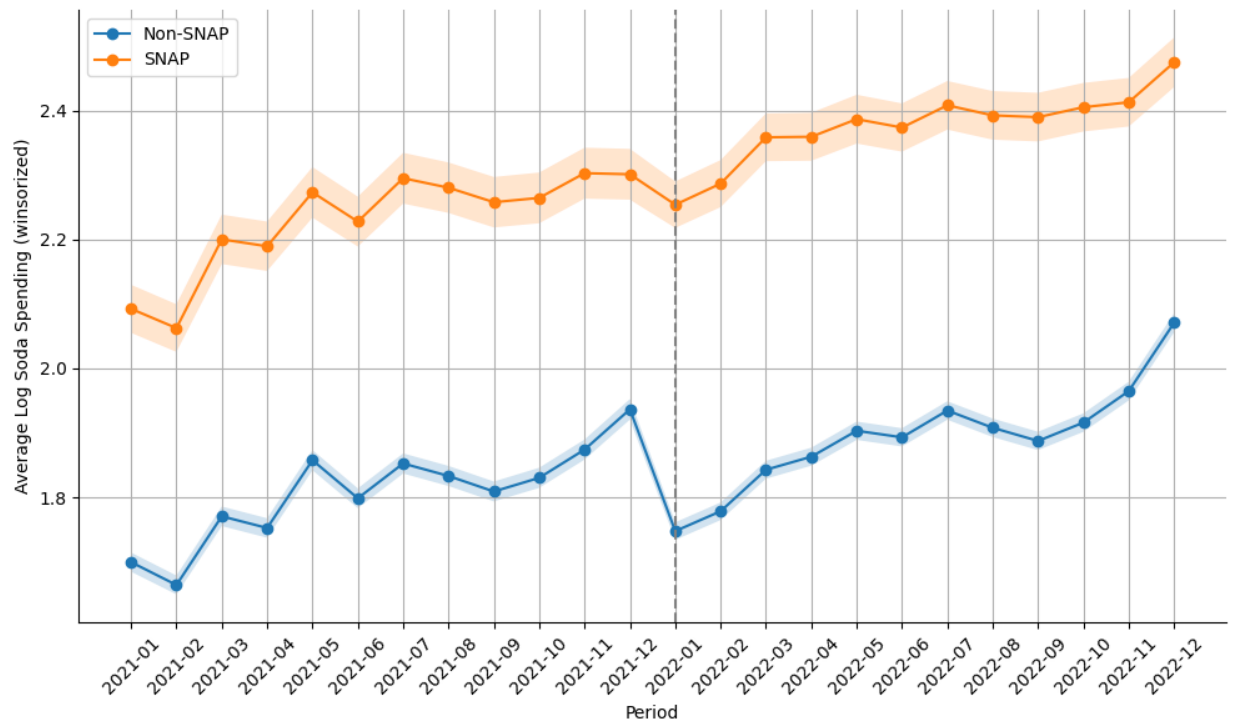
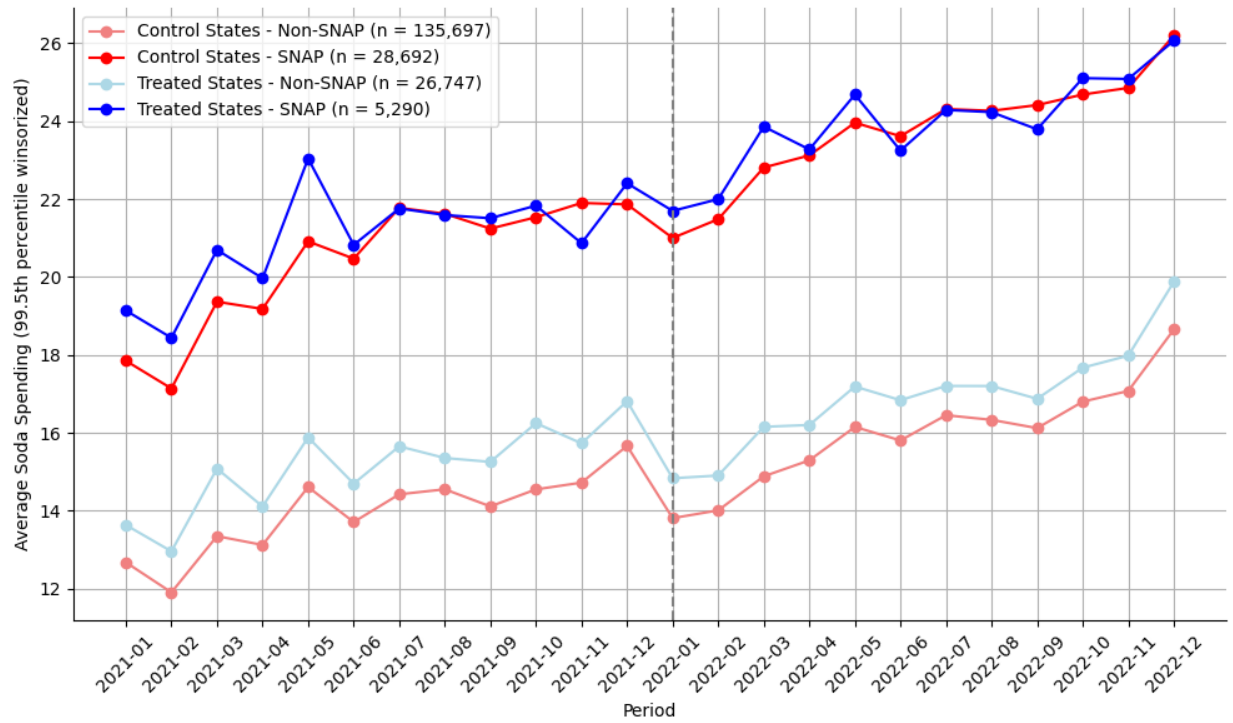


Figure 7: Monthly Average of Soda Spending by SNAP $\times$ state

(a) level



(b) log

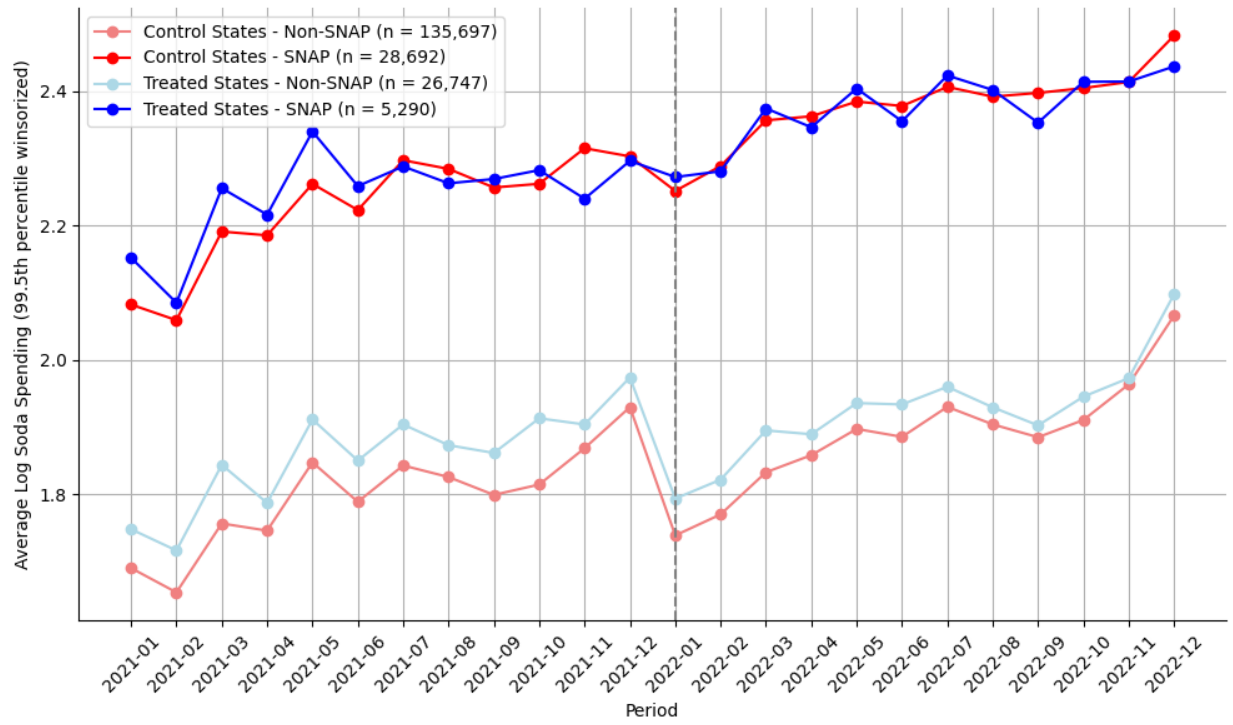


Figure 8: Proportion of Total Spending on Food

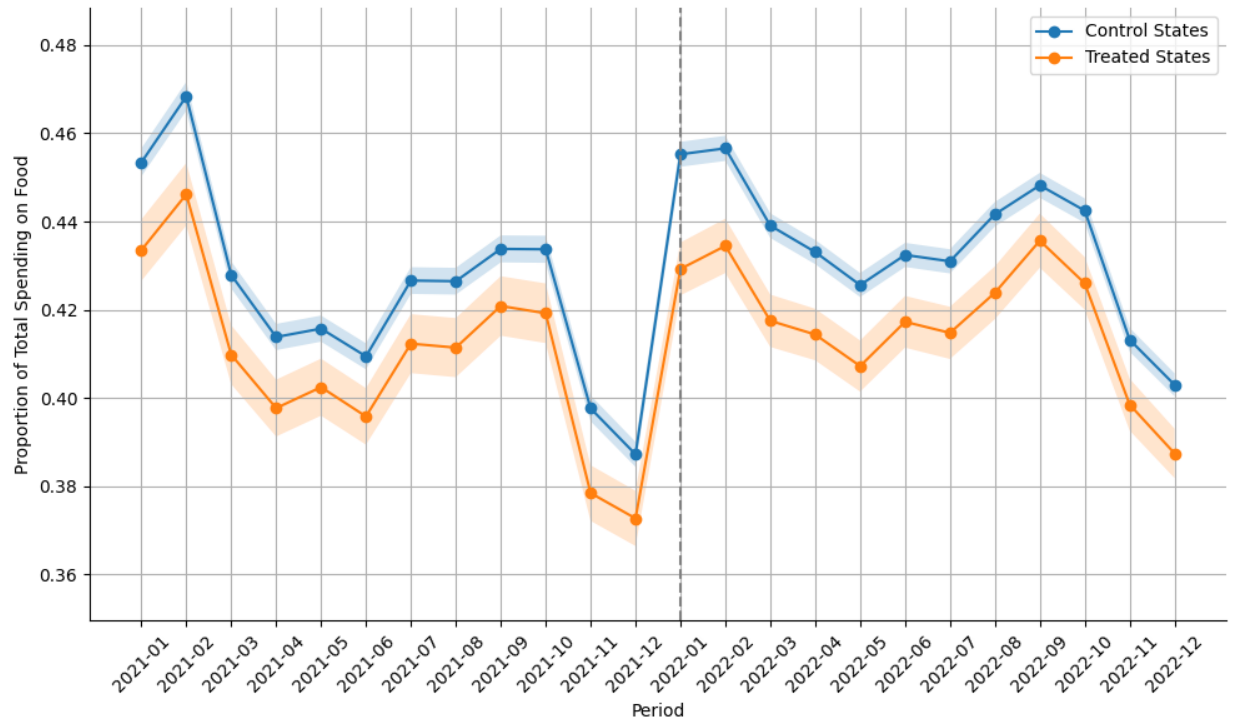


Figure 9: Proportion of Food Spending on EBT

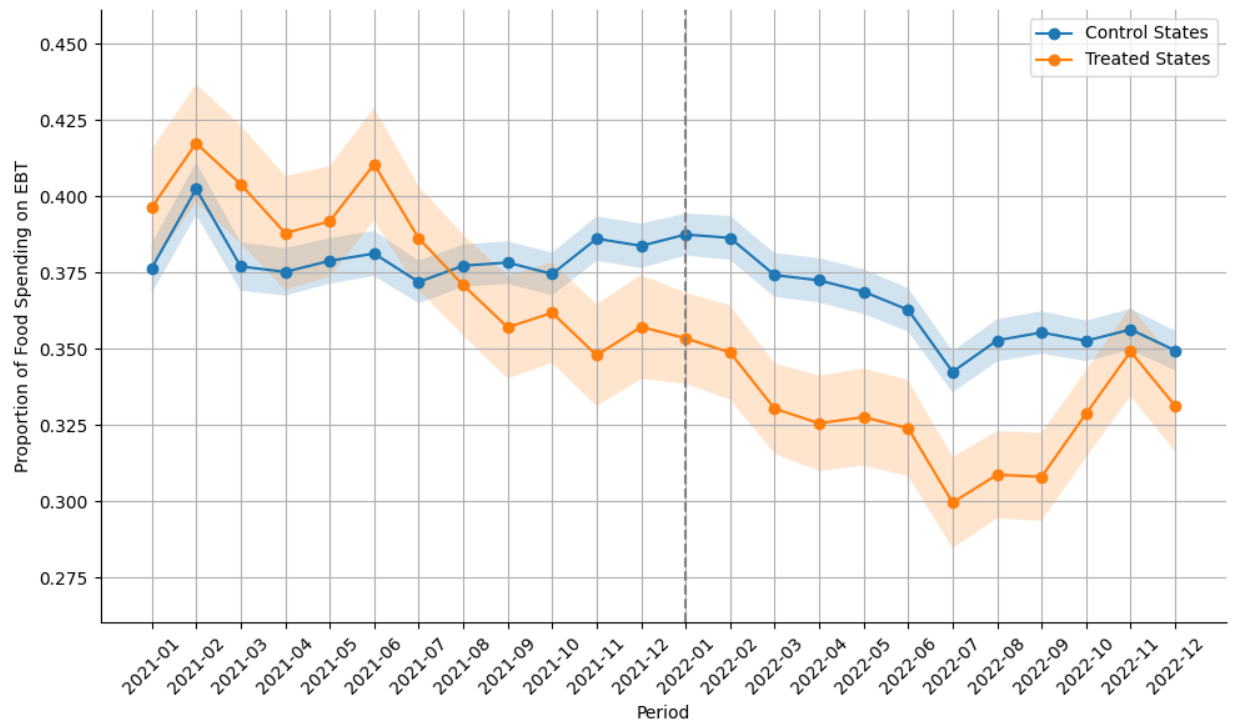


Figure 10: Proportion of Food Spending on Soda

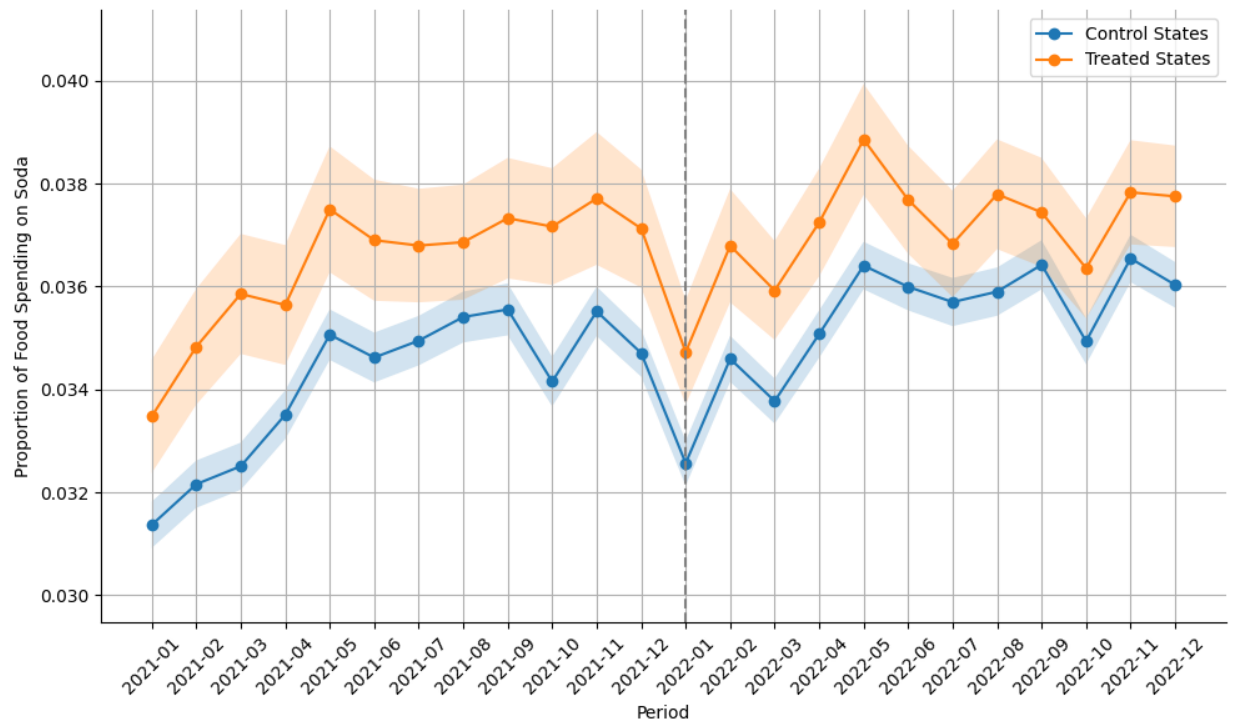


Figure 11: Proportion of EBT Spending on Soda

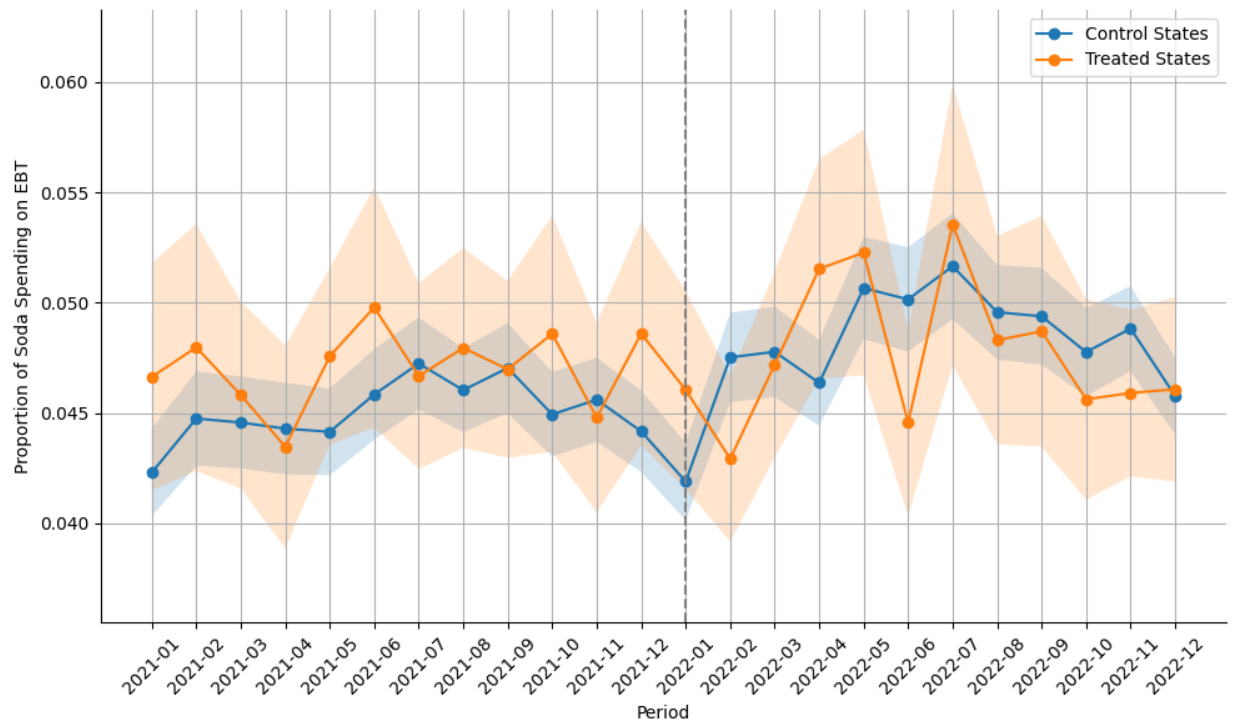


Figure 12: Proportion of Soda Spending on EBT

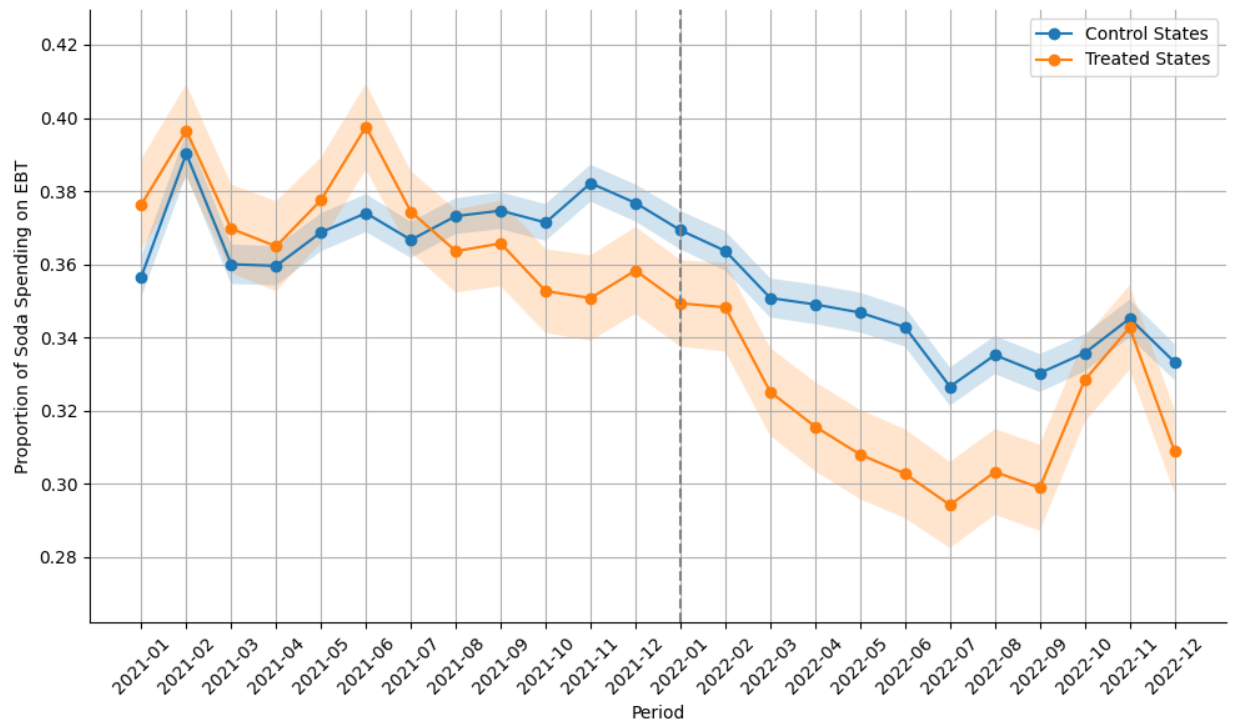
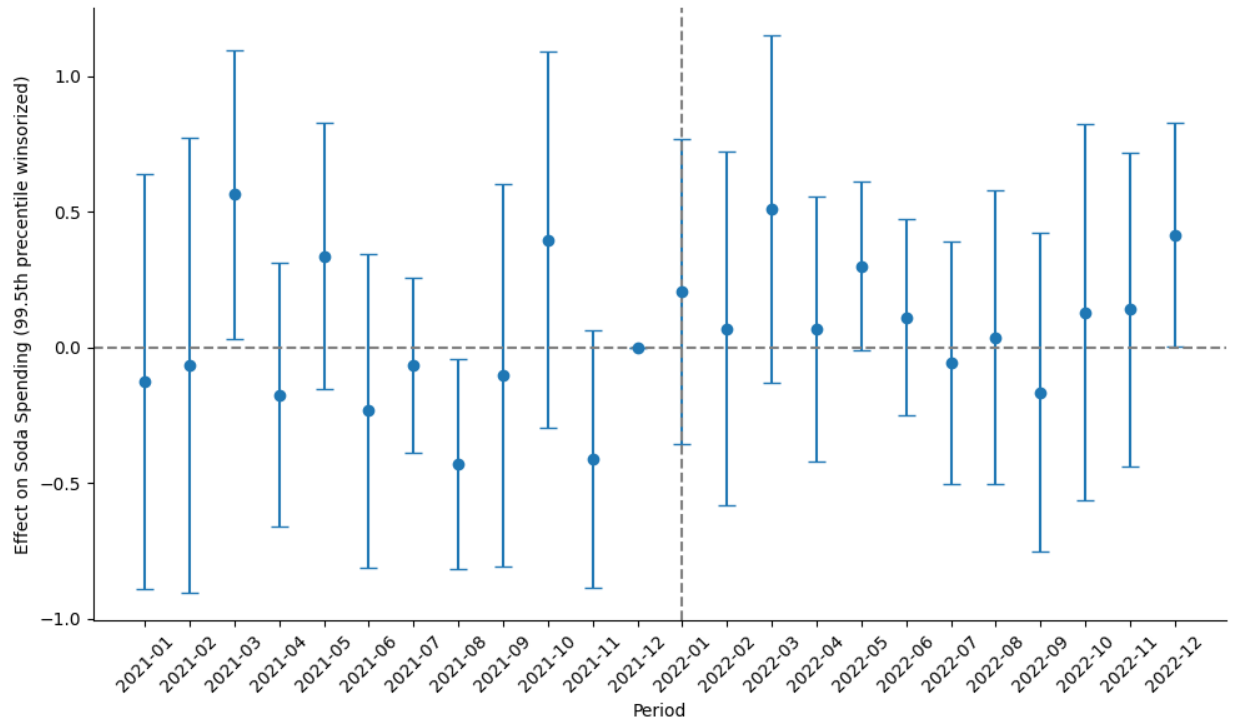


Figure 13: Event Study - TWFE

(a) level



(b) log

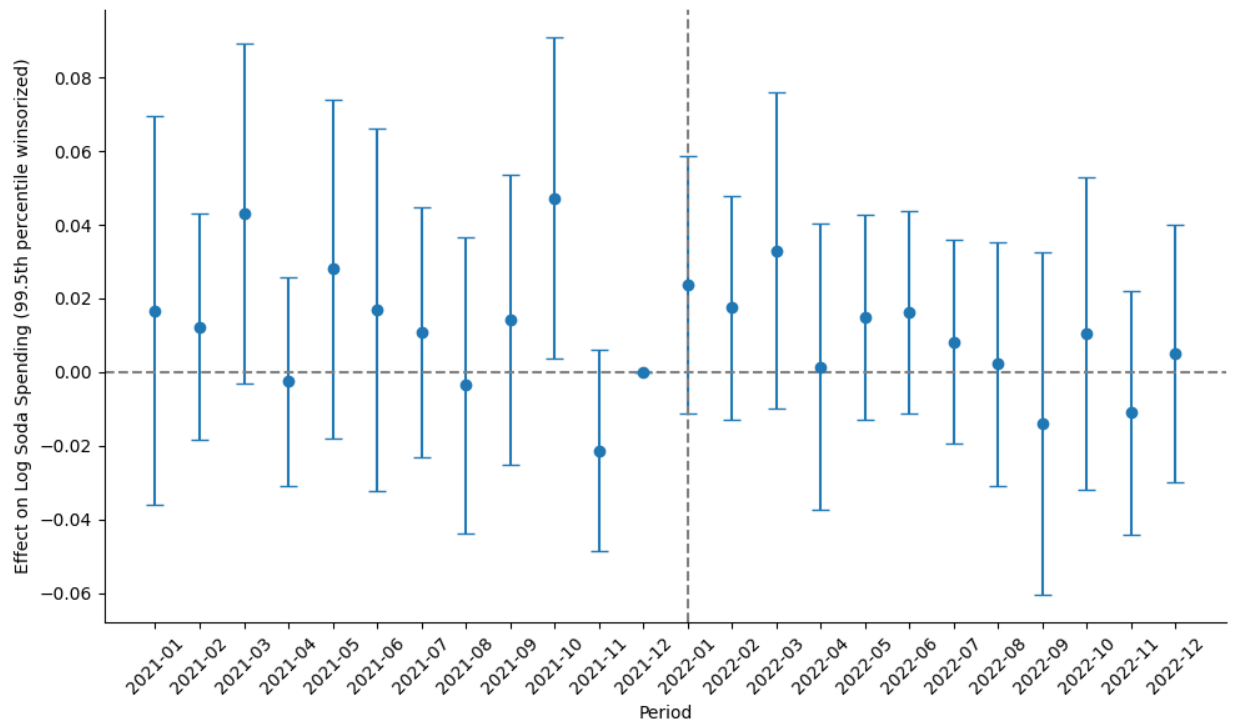


Figure 14: Event Study - TWFE (Treatment Assignment = SNAP/Non-SNAP) within treated states

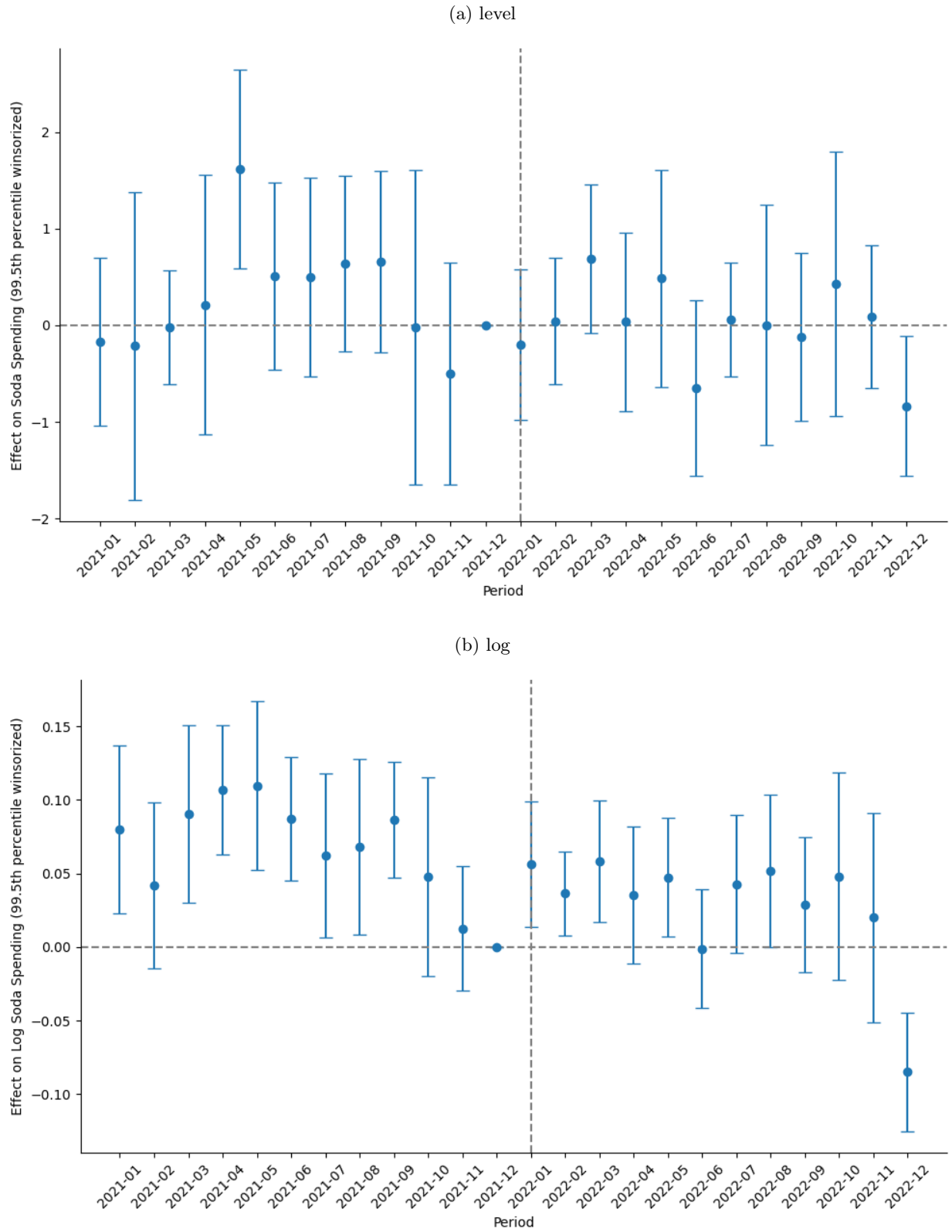


Figure 15: Event Study - DDD

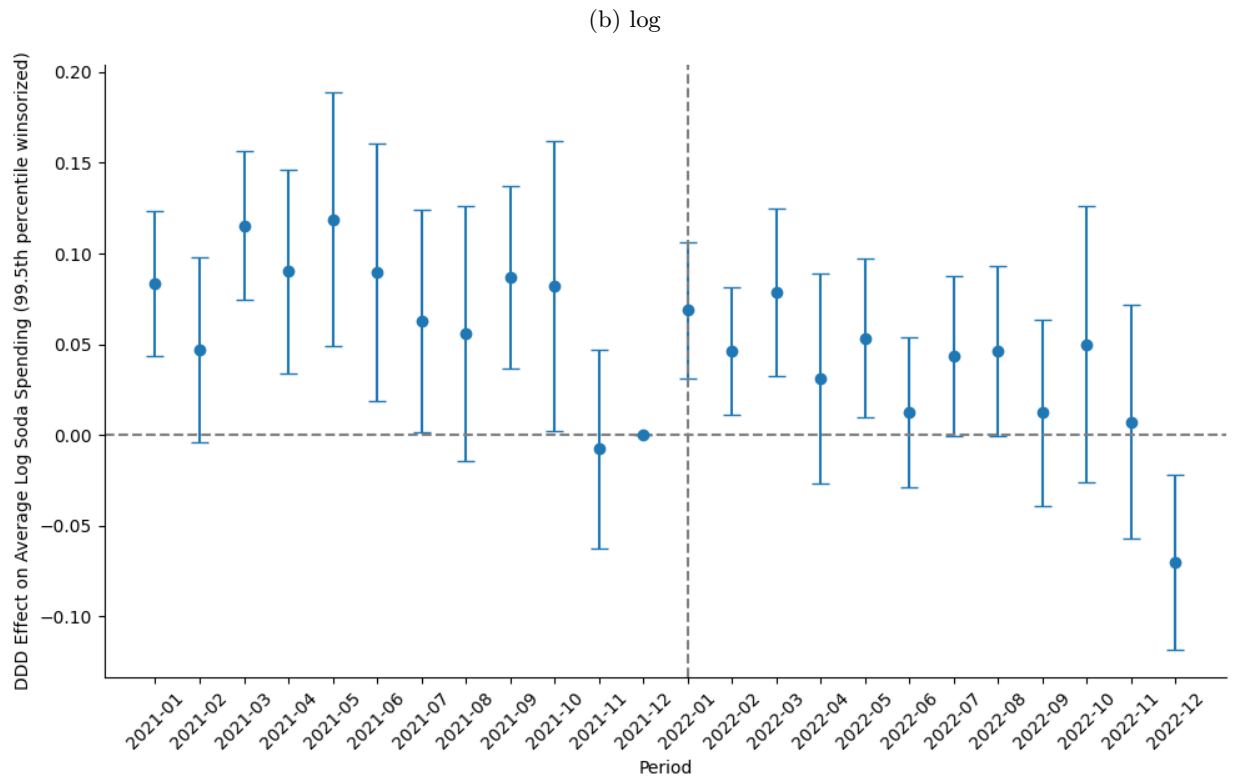
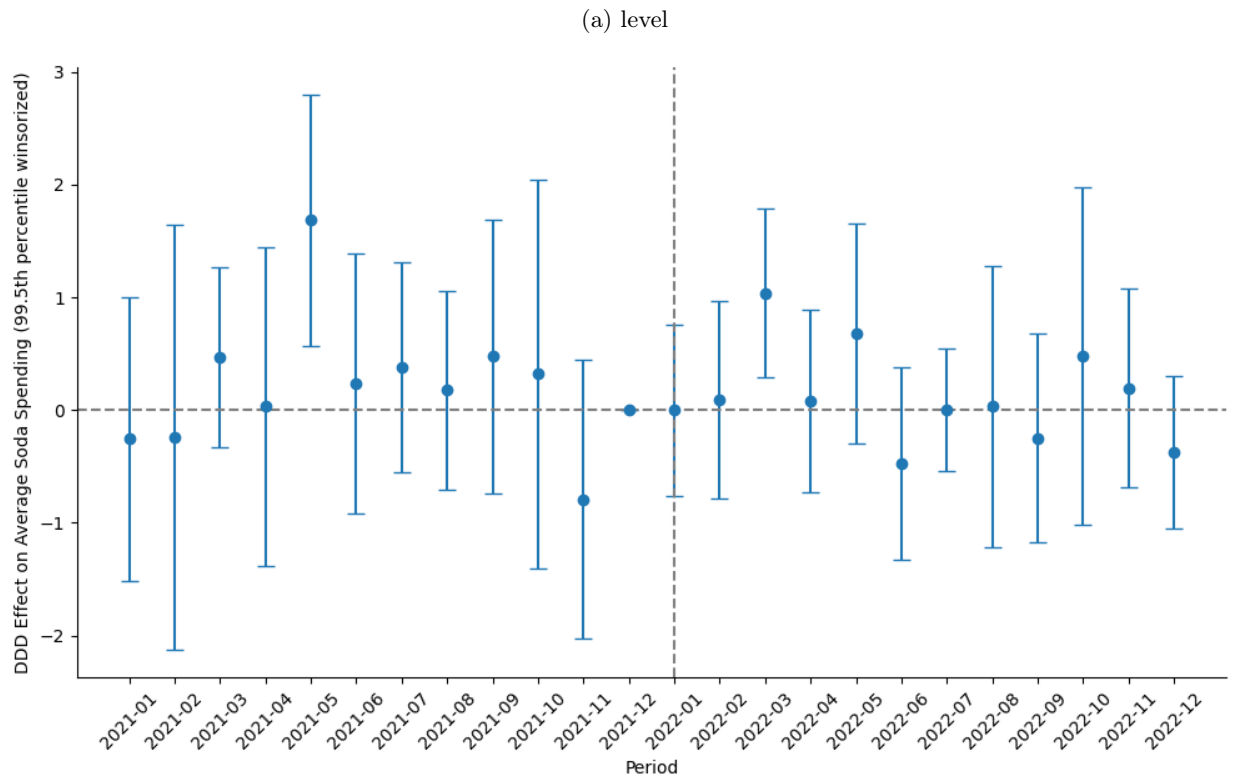


Table 7: Model Estimates

Term	TWFE	TWFE Log	TWFE	TWFE Log	DDD	DDD Log
Post	1.9770*** (0.0662)	0.0857*** (0.0047)	2.1876*** (0.1540)	0.0866*** (0.0115)	2.0059*** (0.0603)	0.0858*** (0.0043)
State $\times$ Post	0.1726 (0.2000)	-0.0045 (0.0120)				
SNAP $\times$ Post			-0.2646 (0.5020)	-0.0379*** (0.0138)		
State $\times$ SNAP $\times$ Post					-0.0828 (0.5628)	-0.0371*** (0.0135)
N	3,592,251	3,592,251	581,327	581,327	3,592,251	3,592,251
R^2	0.0033	0.0013	0.0036	0.0013	0.0033	0.0014
User FE	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	No	No	No	No	No	No
SE Cluster	state	state	state	state	state	state